Midterm practice 2

These are midterm practice problems, not for submission. This is not a "practice midterm" — the number of problems here is greater than the number of problems that will be on the actual midterm.

Practice 2.1. For each statement in the list below: explain whether the statement is true or false, and prove your answer.

Wherever the symbol X appears, it denotes a metric space. Unless otherwise stated, assume the usual distance function on $\mathbb{R}$.

(1) Every nonempty compact subset of $\mathbb{R}$ has at least one limit point.

(2) Every sequence in a compact metric space converges.

(3) If $E \subset X$ and (x_n) is a sequence in E which converges in X, then (x_n) converges in E.

(4) If $E \subset X$ and (x_n) is a sequence in E, then (x_n) is Cauchy in E if and only if it is Cauchy in X.

(5) If (x_n) is a sequence in an a complete metric space with $d(x_{n+1}, x_n) \leq \frac{1}{n}$ then (x_n) converges.

(6) Given any sequence of vectors (x_n) in $\mathbb{R}^k$, with all $x_n \neq (0, \ldots, 0)$, the sequence $\left(\frac{x_n}{\|x_n\|}\right)$ has a convergent subsequence.

(7) $\sum_{n=1}^{\infty} n^2(-3)^n \frac{1}{n!}$ converges absolutely.

(8) If $\sum_{n=1}^{\infty} x_n$ converges, then for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|\sum_{k=1}^{\infty} x_{k+n}| < \epsilon$ for all $n \geq N$.

(9) If X is complete, every closed subset of X is complete.

(10) If $E \subset \mathbb{R}$ is closed and uncountable, it must contain some open interval (a, b).

Practice 2.2. Let $f : [0, \infty)^2 \to [0, \infty)^2$ be the function given by

$$f(x, y) = \left(\sqrt{xy}, \quad \frac{x+y}{2}\right).$$

Given a point $(x_0, y_0) \in [0, \infty)^2$, create a sequence in $\mathbb{R}^2$ by iteratively applying the function f to it, i.e. $(x_{n+1}, y_{n+1}) = f(x_n, y_n)$.

(1) Prove that for any nonnegative real numbers a and b, we have $\sqrt{ab} \leq (a+b)/2$. [1] Conclude that $x_n \leq y_n$ for all $n \geq 1$.

(2) Prove that for any starting point (x_0, y_0), the real sequence (x_n) is monotonically increasing for $n \geq 1$ and the sequence (y_n) is monotonically decreasing for $n \geq 1$.

(3) Prove that for any starting point (x_0, y_0), the sequence $((x_n, y_n))$ in $\mathbb{R}^2$ converges to a limit of the form (L, L) for some real number $L \geq 0$. You may use without proof that f is continuous, but make sure to explain if/where you use this fact.

[1] The expression on the left is called the *geometric mean* of a and b, whereas the expression on the right is the *arithmetic mean*. This says that the geometric mean is always less than or equal to the arithmetic mean of two nonnegative real numbers.

Practice 2.3. Let $f : [0,1] \to \mathbb{R}$ be a function such that $\lim_{x \to a} f(x) = 0$ for all $a \in [0,1]$.

(1) Show that the set of $a \in [0,1]$ such that $|f(a)| > b$ is finite for any $b > 0$.

(2) Show that the set of points where f is discontinuous is at most countable.